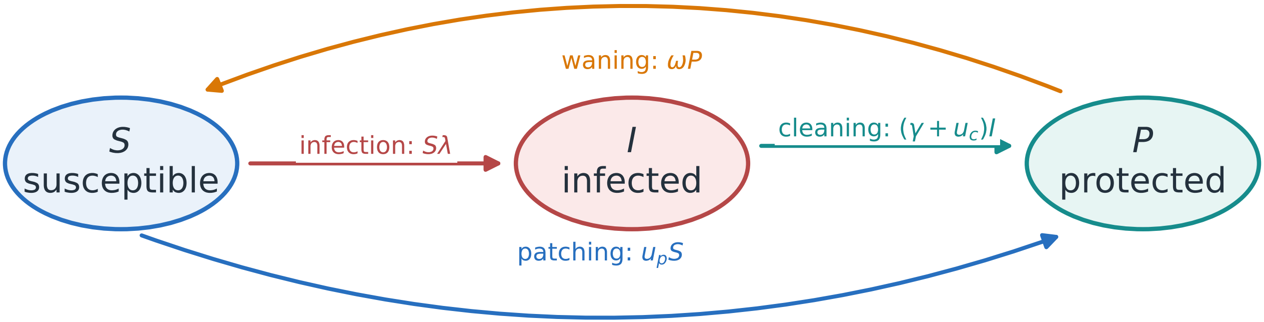
Canonical SIPS state transitions

State order is $[S, I, P]$ and $S + I + P = 1$; every flow transfers mass between compartments.



Aggregate pressure
$\lambda(t) = \beta I(t)$
or degree mixing $\lambda_k = s_k k \sum_{\ell} M_{k\ell}\, \iota_\ell x_\ell$

Node-indexed pressure
$\lambda_i = \beta_i s_i \sum_j A_{ij}\, \iota_j I_j$
node-specific $s_i, \iota_i, \gamma_i, \omega_i$
and node-specific control bounds

The invariant is a modeling check, not an extra loss term in the ODE solver.